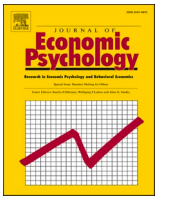
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Brief Report



Between money and speculative asset: The role of financial literacy on the perception towards Bitcoin in Italy☆

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ABSTRACT

With Bitcoin at the forefront, cryptocurrencies are gaining traction as an alternative asset investment, particularly among young investors. Although most of the empirical evidence has shown that it could not be defined as a currency, some Bitcoin users argue the opposite. This paper analyzes the factors influencing the perception of Bitcoin, i.e., whether it is a currency or an asset, with a focus on financial literacy among a subject pool of university students in Italy. The results show that, after controlling for several individual characteristics such as behavioral biases, personal attitudes, psychological traits, and socio-demographic information, this cryptocurrency is considered more than just an asset, and thus it could replace currency, among subjects with lower financial literacy, higher knowledge of Bitcoin, and those who do not trust the banking system. In contrast, Bitcoin is considered a speculative asset among those individuals with higher financial literacy. In line with the recent evidence that cryptocurrencies are mostly owned by young investors, results indicate the importance of increasing the level of financial education among them.

1. Introduction

In recent years, cryptocurrencies have become a widely used financial instrument. In particular, Bitcoin has gained significant attention as an alternative asset, reaching a trillion dollars market capitalization (CNBC, 2021).

Although the increasing interest in Bitcoin as a digital currency, which was created in 2008 by an unknown person with the pseudonym of Satoshi Nakamoto, empirical works mainly argue that cryptocurrencies should be seen as investment instruments (assets) rather than alternative currencies (money). Analyzing transaction data, Baur et al. (2018) point out that Bitcoin is mainly used as a speculative asset and not as an alternative currency or a medium of exchange. Also, Söderberg (2018) points out that crypto assets cannot be classified as money. The same conclusion is supported by Griffin and Shams (2020) who argue that, despite cryptocurrencies being designed to bypass traditional banking systems, these do not remove the need for external surveillance, monitoring, and a regulatory framework, typical of centralized currency systems. Despite the potential financial bubble characterizing crypto assets (Caferra et al., 2021; Vidal-Tomás et al., 2019), some studies support the idea that Bitcoin can act as a store of value. Blau et al. (2021) showed that Bitcoin has shown potential as a hedge against inflation since its price variation tends to lead to changes in the expected inflation (for a review of investors' behavior in cryptocurrency markets see Almeida and Gonçalves, 2023).

In the vein of Bitcoin as money, an important and recent case study is related to the experience of El Salvador, which is the first country to experiment with the adoption of Bitcoin as a legal means of exchange (The Economist, 2021). Starting on 7th September

☆ Data and code to replicate the analysis can be accessed from the OSF repository at the link: https://osf.io/vjdyn/.

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2021, through the legislative measure known as the "Bitcoin Law", the Country's Government imposed that Bitcoin must be accepted as a means of payment for taxes and outstanding debts for citizens, and as a medium of exchange for all transactions among firms, which are obliged to accept bitcoin through the "Chivo Wallet" app. Despite the Government's efforts and incentives in making Bitcoin popular and commonly usable (i.e., $30 in free Bitcoin for each Chivo Wallet download, as well as discounts on gasoline paid for in Bitcoin), evidence from a quasi-natural experiment across a representative sample of 1.800 Salvadorans showed that the adoption of Bitcoin as a medium of exchange remained quite limited, and the usage of Chivo is not widespread, considering that only 20 percent of businesses comply with the law to accept bitcoin, 5 percent of sales are made through Chivo Wallet, and 88 % percent of firms convert bitcoin to dollars (Alvarez et al., 2023).

All these works are based on empirical transaction data, the literature lacks a paper discussing an important behavioral aspect of this debate, namely whether and why people perceive Bitcoin as money or as an asset. Some survey studies exist about the adoption of cryptocurrencies, mainly arguing that, although cryptocurrencies are increasing in popularity across all age groups (Hasso et al., 2019), this asset class is mostly owned by young people (Steinmetz et al., 2021; see Sousa et al., 2022, for a systematic review on the adoption of cryptocurrencies). Moreover, according to the evidence of Kumar (2009), a relationship exists between stock investment decisions and individuals' gambling preferences. Considering that young people tend to be less risk-adverse than adults and that cryptocurrencies have often been associated with lottery investments, it is worth investigating how young investors perceive the role of Bitcoin.

However, most of these studies do not investigate the perception of Bitcoin as a money or asset. Therefore, conducting an original survey on a subject pool of 682 university students belonging to Italy, a country characterized by a low level of financial literacy (see Klapper et al., 2015, World Bank, for an international comparison of financial literacy levels), this paper addresses this research gap by providing an empirical analysis of the determinants of the perception towards Bitcoin as a money or asset, with a focus on the role of financial literacy.

2. Literature review on the Bitcoin perception

Despite the relatively limited literature about the psychological and behavioral variables influencing the perception of Bitcoin as money or asset, some recent evidence regarding the intentions of individuals toward the adoption of this cryptocurrency may allow us to infer some possible determinants. For instance, using the extended theory of planned behavior, Anser et al. (2020) argue that social media usage is a significant driver of the intention to adopt Bitcoin. Based on empirical findings from a Chinese survey, the authors found out the usage of social media is positively associated with users' intentions toward adopting Bitcoin through their attitudes, subjective norms, and perceived behavioral control. Furthermore, intentions seem to be linked to actual Bitcoin adoption behavior, with perceived risk moderating this relationship. Employing the same methodology, Kim (2021) analyzes the roles of general consumers' attitudes toward money in predicting behavioral intention to use Bitcoin for daily transactions. Based on 395 samples from general consumers in the United States who are interested in Bitcoin, the author evidenced how power prestige, retention time, and distrust may significantly predict the behavioral intention to use Bitcoin at the individual level. In turn, power prestige, retention time, distrust, quality, and anxiety are influenced by perceived behavioral control, subjective norms, and financial self-efficacy. This evidence is confirmed by Bommer et al. (2023), who provided a *meta*-analysis about the intention to use cryptocurrency over 42 studies, referring to 9482 subjects across emerging and developed economies. In fact, the authors argue that personal attitudes, performance expectations, price value, and social influence significantly influence the intent to use cryptocurrencies, as well as other factors such as facilitating conditions, effort expectancy, and perceived behavioral control.

Another strand of literature focuses on the role of trust towards the usage of Bitcoin since the latter could increase it. In this vein, Ooi et al. (2021) confirmed this relationship by investigating the role of perceived security and perceived trust in Bitcoin usage. The authors provided evidence that Bitcoin users' perceived trust in Bitcoin mediates the relationship between perceived security and Bitcoin use. In fact, the importance of addressing usability aspects for enhanced user acceptance and trust in Bitcoin is deepened by Alshamsi and Andras (2019), who compared the difference in usability and security between credit and debit cards and Bitcoin from the users' perspective. Results from human-computer interaction showed that credit and debit cards are still overall more usable and secure than Bitcoin, due to differences in learnability, efficiency, help, security, and satisfaction. More recently, Dabbous et al. (2022) focused on the possible influence of risk on the adoption of cryptocurrencies. The author showed evidence from a behavioral perspective of the factors that influence individuals' adoption of cryptocurrencies for financial transactions within a high-risk context. Based on a sample of 255 subjects in Lebanon, the study finds that the perceived risk is lower across individuals who are more aware of financial technology. Moreover, social influence may have a significant role in reducing the perceived risk of cryptocurrencies, while this latter is positively related to both subjective risk aversion and the presence of regulatory support. In the context of market risk, Kliber et al. (2019) explore the role of Bitcoin as a hedge, diversifier, or haven in various stock markets, using daily market data over the period 2014–2017, considering economic situations in five diverse countries (Japan, Venezuela, China, Estonia, and Sweden). Employing a multivariate stochastic volatility model with dynamic conditional correlation, the study showed that, on one hand, Bitcoin emerged as a weak hedge across all markets, even though on the other its role may be heterogeneous across countries. In fact, Bitcoin seems to act as a haven only in Venezuela and bolivars, while it works as a diversifier in Japan and China, and finally, it showed features of a weak hedge in Sweden and Estonia. Also taking as reference the experience of El Salvador, previously discussed, the usage of Bitcoin remains limited with respect to the classical electronic payments, such as credit and debit cards.

3. Research hypotheses

Starting from the whitepaper by Nakamoto (2008), Bitcoin has been born as a decentralized alternative to the actual financial and banking system. Despite this, most of the empirical works fail to define Bitcoin as a new currency. In fact, according to transaction data, it does not fit the functional definition of currency, which must act as I) means of payment; II) unit of account; and III) store of value (Jevons, 1875). Rather, it seems to follow the stylized facts of classical financial assets (Caferra et al., 2022). However, these stylized facts and the definition of money could be reasonably known only by individuals with a deeper financial literacy. As discussed by Darriet et al. (2020), financial literacy plays a significant role in reducing money illusion bias, which in turn may lead to wrong financial behaviors. In this case, it is reasonable to expect that people with a higher financial literacy should evaluate Bitcoin as a speculative asset only, while the opposite is expected for those with lower financial literacy. The first hypothesis (*H1*) is formulated accordingly.

Hypothesis 1 Higher (lower) financial literacy is positively related to the perception of Bitcoin as an asset (money).

Nevertheless, the increasing popularity and mainstream adoption of Bitcoin could not be related to its initial aim. In fact, it seems that people tend to invest in Bitcoin mainly to get a positive return in time (i.e, herding effect), typical of the individual's behavior generating financial bubbles (see Caferra et al., 2021, for a comparison between the crypto's 2017 bubble and Dotcom bubble of 2000). In the context of financial bubbles, Vidal-Tomás et al. (2019) point out that the herding phenomenon does not refer to Bitcoin only, and the smallest digital currencies are herding with the largest ones. Since these dynamics are mainly driven by the possibility of gaining an economic return, it is reasonable to assume that Bitcoin could be considered as an alternative to traditional currencies only for those who know about Bitcoin more in-depth, being more in line with the Nakamoto's ideas rather than positive return only. Accordingly, we formulate the following hypothesis (H2).

Hypothesis 2 Higher (lower) knowledge about Bitcoin is positively related to the perception of Bitcoin as money (asset).

4. Methodology

In line with the recent evidence that cryptocurrencies are mostly owned by young investors (Steinmetz et al., 2021), we have surveyed a group of 682 Italian university students. The survey was conducted via Microsoft Forms and spread out on social media (Instagram) in February 2023. The structure of the survey is reported in the Online Appendix.

The dependent variable is obtained from the survey question "On *a scale from 1 to 5, where 1 is not at all, and 5 is definitely, how much do you think Bitcoin can replace the current currencies in circulation?"*. In order to test the complementary hypothesis, namely Bitcoin is perceived as an asset, another dependent variable has been proposed. It is a dummy variable taking the value of 1 if respondents declared that Bitcoin is an asset.

The first main regressor is a measure of financial literacy. The methodology used to construct it has been taken from Hsu (2022), which refers to Calcagno and Monticone (2015). It is proxied by respondents' answers regarding (I) inflation, (II) interest, and (III) diversification. As in Bannier et al. (2019), we also control for the perceived financial literacy, which level has been obtained from the question "*What level of financial education do you think you have*?" on a 10-point Likert scale. It is important to control for this variable since, according to Balasubramnian and Sargent (2020), an inflated perception of the subjective financial literacy level may negatively affect financial decisions, leading to miss mortgage payments and poor banking accounts. This evidence is strengthened by Bucciol et al. (2021) who, testing the effect of a short finance lecture on a group of elderly people, showed that subjects could increase their confidence without a corresponding increase in competence after such a short course, implying a potential problem of overconfidence bias.

The other main regressor refers to the knowledge about Bitcoin, which is proxied by the *Bitcoin literacy index,* obtained by re-proposing the same survey questions of Bannier et al. (2019), who introduced this concept in the literature. It results in an ordinal index measuring the number of correct answers to the 6 Bitcoin literacy questions, which are reported in section 1 of Online Appendix. We then also control for Bitcoin usage, obtaining the respondent's information on whether he or she has ever invested in or used Bitcoin to make payments. All these variables have been included in the estimation of the baseline model.

In addition to this baseline model, we then included behavioral biases and personal attitudes as control variables that may affect the perception of Bitcoin as money or an asset. First, drawing on Hsu (2022), we control for overconfidence, self-enhancement, self-protection, and mental accounting bias. Moreover, we include some personal attitudes towards the financial system and social preferences, such as financial stress; generalized trust; trust in institutions, banks, and the central bank; political orientation and participation; social capital; tax morality; climate change, and informatic concern.

Finally, a full model that considers personality traits and socio-demographic information is proposed. In fact, according to Tang and Baker (2016), the importance of psychological traits can be relevant factors explaining financial behavior differences. Therefore, following Bannier et al. (2019) and Brown and Taylor (2014), the big five personality traits are also included (openness, conscientiousness, extraversion, agreeableness, and neuroticism), as well as the socio-demographic information regarding gender, age, employment status, and educational attainment.

5. Results

Given the ordinal nature of the dependent variable, Table 1 reports the ordered probit estimates of the determinants of Bitcoin as a possible substitute for actual money. As it is visible, the hypotheses seem to be confirmed: the higher the financial literacy, the lower the perception of Bitcoin as money (H1). On the contrary, the probability of stating that Bitcoin can substitute money is significantly

Table 1
Ordered probit estimates Dependent variable "Bitcoin substitutes money" (1–5).

	(1) Baseline		(2) Biases and attitudes		(3) Full model	
	Coeff.	Averag ME †	Coeff.	Average ME †	Coeff.	Average ME †
Financial literacy	−0.141***	−0.009	−0.111**	−0.006	−0.078*	−0.004
	(0.034)		(0.035)		(0.037)	
BTC literacy index	0.105**	0.006	0.099*	0.006	0.106**	0.006
	(0.038)		(0.039)		(0.040)	
Invested in BTC	0.499***	0.030	0.437***	0.024	0.487***	0.026
	(0.102)		(0.105)		(0.109)	
Paid in BTC	0.632***	0.038	0.500**	0.028	0.464**	0.025
	(0.169)		(0.173)		(0.178)	
Perceived financial literacy	−0.035	−0.002	−0.007	−0.000	0.024	0.001
	(0.028)		(0.029)		(0.031)	
Behavioral biases						
Over confidence			0.008	0.000	−0.021	−0.001
			(0.118)		(0.119)	
Self enhancement			0.533	0.030	0.543	0.029
			(0.284)		(0.290)	
Self protection			−0.150	−0.008	−0.193	−0.010
			(0.106)		(0.109)	
Mental accounting			0.133	0.007	0.171	0.009
			(0.144)		(0.146)	
Attitudes						
Financial stress			0.057	0.003	0.081	0.004
			(0.099)		(0.101)	
Social capital			−0.272*	−0.015	−0.281*	−0.015
			(0.106)		(0.108)	
Voted in elections			−0.011	−0.001	−0.004	0.000
			(0.123)		(0.128)	
Trust in institutions			−0.038	−0.002	−0.030	−0.002
			(0.054)		(0.055)	
Trust in banks			−0.106	−0.006	−0.108	−0.006
			(0.064)		(0.065)	
Trust in ECB			−0.143*	−0.008	−0.119*	−0.006
			(0.060)		(0.061)	
Generalized trust			−0.010	−0.001	−0.007	−0.000
			(0.056)		(0.058)	
Tax morale			−0.069	0.004	−0.078	0.004
			(0.050)		(0.051)	
Political orientation			−0.049	−0.003	−0.040	−0.002
			(0.053)		(0.054)	
Climate change concern			−0.000	−0.000	−0.006	−0.000
			(0.048)		(0.050)	
Informatic concern			0.081	0.005	0.075	0.004
			(0.045)		(0.045)	
Personality traits						
Openness					0.016	0.000
					(0.059)	
Conscientious					−0.061	−0.003
					(0.065)	
Extraversion					−0.030	−0.002
					(0.041)	
Agreeableness					−0.007	−0.000
					(0.056)	
Neuroticism					−0.060	−0.003
					(0.048)	
Socio demographics						
Female					0.336**	0.018
					(0.114)	
Age					−0.000	−0.000
					(0.013)	
Neet					0.921*	0.089
					(0.457)	
Unemployed					0.545	0.0.039
					(0.421)	
Self-employed					0.772**	0.067

(*continued on next page*)

Table 1 (*continued*)

	(1) Baseline		(2) Biases and attitudes		(3) Full model	
	Coeff.	Averag ME †	Coeff.	Average ME †	Coeff.	Average ME †
					(0.299)	
Employee					0.186	0.010
					(0.174)	
Worker/student					0.076	0.004
					(0.103)	
Education: high school diploma					−0.079	−0.006
					(0.501)	
Bachelor's Degree					−0.271	−0.015
					(0.507)	
Master's Degree					−0.262	−0.015
					(0.520)	
PhD					0.631	0.069
					(0.619)	
Pseudo R-squared	0.050		0.075		0.093	
Observations	682		682		682	

Standard errors in parentheses. * p <.05 ** p <.01 *** p <.001.
The marginal effects are computed at the highest level of the dependent variable (5).

higher among individuals who show a deeper knowledge of Bitcoin and those who have invested or made payments in Bitcoin. These results are consistent across all three proposed specifications. From the second and third models, it is visible that the effect of behavioral bias is not significantly related to the perception of Bitcoin. Among attitudes, people who trust more the European Central Bank do not consider Bitcoin as a potential substitute for money, and vice versa. Also, social capital significantly affects the perception towards Bitcoin: those individuals involved in voluntary associations seem to not value Bitcoin as a substitute for money. The personality traits do not significantly affect the perception of Bitcoin as money. Finally, among the socio-demographic information, Bitcoin seems to be seen as money significantly more among females, NEET, and self-employed workers.

To test the complementary side of the perception, namely, Bitcoin is considered an asset, Table 2 reports the probit estimates of the probability of stating that Bitcoin is just a speculative asset. Estimates are overall coherent with the previous ones, confirming both research hypotheses. As one can see from the first column, coherently with the evidence in Table 1, higher financial literacy is positively related to the perception of Bitcoin as an asset. On the contrary, Bitcoin is perceived to be just an asset significantly less among respondents who have a higher general knowledge about Bitcoin and those who have used it as an investment and/or a means of payment. The latter evidence is key: people who invest in Bitcoin do not see it as a speculative investment, rather they expect Bitcoin to substitute money in the future. Hence, Bitcoin users consider it more than a simple asset. Also in this context, people who trust more the central banking system perceive Bitcoin as an asset rather than a substitute for money. Considering that Bitcoin may be used to make illegal payments, subjects who do not justify tax evasion coherently see it as a speculative asset rather than a means of payment.

6. Discussion

Although the results analyzed so far provide novel insights about behavioral factors influencing the perception of Bitcoin, it is important to specify that these findings are hardly generalizable. Indeed, the data only refer to an Italian sample, which may be characterized by specific cultural factors or fixed effects that may differ for other countries. In particular, it is well-known that the Italian population exhibits a particularly scarce level of financial literacy (see, for instance, D'Alessio et al., 2020; Klapper and Lusardi, 2019), the lowest across major advanced economies (SP, 2015). Therefore, the results provided in this paper could be overall influenced by this factor. Despite this referring to a limitation of the study, it might be worthwhile replicating it in a context with a more virtuous financial education level, to understand whether some significant differences arise in the results, and if other factors may turn out to mediate the results presented in this paper.

Another important factor to be considered, specific to this study, is the timeframe during which the research survey has been conducted. In particular, this period is characterized by two significant events in the cryptocurrency environment, namely, the collapse of the "Terra-Luna" crypto and the FTX platform (Cointelegraph, 2022). Accordingly, as argued by Vidal-Tomás et al. (2023), the collapse of the Terra-Luna has been the trigger for the FTX's crash and, hence, the role of beliefs in the cryptocurrency environment becomes crucial, since users may show very strong beliefs regarding any kind of event related to this world.

7. Concluding remarks

Although it is increasing in popularity, Bitcoin cannot be defined as money according to most of the literature. However, it is essential to understand what leads people to consider it as an asset or an alternative to the actual currency system. To this aim, we conducted a survey to elicit young individuals' perception of Bitcoin as money or an asset, focusing on the role of financial literacy. Results showed that subjects with a higher financial literacy recognize that Bitcoin cannot substitute money, rather it is a speculative asset. On the opposite, only individuals with a deeper knowledge of Bitcoin consider it as a currency. Although this evidence, Bitcoin

Table 2
Probit estimates Dependent variable "Bitcoin is an asset" (dummy).

	(1) Baseline		(2) Biases and attitudes		(3) Full model	
	Coeff.	Average ME †	Coeff.	Average ME †	Coeff.	Average ME †
Financial literacy	0.172***	0.053	0.149**	0.044	0.154**	0.044
	(0.043)		(0.045)		(0.048)	
BTC literacy index	−0.132**	−0.041	−0.114*	−0.034	−0.128*	−0.036
	(0.049)		(0.052)		(0.054)	
Invested in BTC	−0.582***	−0.179	−0.518***	−0.153	−0.600***	−0.171
	(0.126)		(0.131)		(0.139)	
Paid in BTC	−0.784***	−0.242	−0.748**	−0.221	−0.658**	−0.189
	(0.211)		(0.222)		(0.228)	
Perceived financial literacy	0.060	0.019	0.043	0.013	0.051	0.015
	(0.035)		(0.037)		(0.040)	
Behavioral biases						
Over confidence			0.110	0.032	0.112	0.031
			(0.154)		(0.157)	
Self enhancement			0.234	0.069	0.246	0.070
			(0.339)		(0.356)	
Self protection			0.081	0.024	0.088	0.025
			(0.136)		(0.141)	
Mental accounting			−0.032	−0.009	−0.034	−0.010
			(0.193)		(0.198)	
Attitudes						
Financial stress			−0.096	−0.028	−0.095	−0.027
			(0.126)		(0.129)	
Social capital			0.366*	0.108	0.381*	0.110
			(0.145)		(0.150)	
Voted in elections			−0.162	−0.048	−0.247	−0.072
			(0.164)		(0.174)	
Trust in institutions			−0.036	−0.011	−0.034	−0.010
			(0.071)		(0.073)	
Trust in banks			0.081	0.024	0.045	0.013
			(0.082)		(0.085)	
Trust in ECB			0.156*	0.046	0.157*	0.045
			(0.078)		(0.080)	
Generalized trust			−0.009	−0.003	0.002	0.001
			(0.073)		(0.076)	
Tax morale			−0.133*	−0.039	−0.142*	−0.040
			(0.065)		(0.067)	
Political orientation			0.026	0.008	0.025	0.007
			(0.068)		(0.071)	
Climate change concern			−0.013	−0.004	−0.009	−0.003
			(0.063)		(0.065)	
Informatic concern			−0.040	−0.012	−0.023	−0.007
			(0.059)		(0.060)	
Personality traits						
Openness					−0.066	−0.019
					(0.080)	
Conscientious					0.028	0.008
					(0.086)	
Extraversion					−0.080	−0.023
					(0.054)	
Agreeableness					0.137	0.039
					(0.076)	
Neuroticism					0.007	0.002
					(0.065)	
Socio demographics						
Female					−0.152	0.038
					(0.150)	
Age					−0.029	−0.008
					(0.017)	
Neet					0.235	0.085
					(0.561)	
Unemployed					−0.224	−0.068
					(0.499)	
Self-employed					−0.544	−0.174

(*continued on next page*)

Table 2 (*continued*)

	(1) Baseline		(2) Biases and attitudes		(3) Full model	
	Coeff.	Average ME †	Coeff.	Average ME †	Coeff.	Average ME †
					(0.397)	
Employee					−0.089	−0.028
					(0.225)	
Worker/student					0.089	0.024
					(0.138)	
Education: high school diploma					0.712	0.227
					(0.612)	
Bachelor's Degree					0.785	0.246
					(0.621)	
Master's Degree					0.696	0.221
					(0.637)	
PhD					0.423	0.132
					(0.784)	
Constant	−0.268		−0.671		−0.791	
	(0.306)		(0.680)		(1.023)	
Pseudo R-squared	0.111		0.145		0.168	
Observations	682		682		682	

Standard errors in parentheses. * p <.05 ** p<.01 *** p <.001.

users who invest in this crypto do not see it as a speculative investment only, rather they expect Bitcoin to substitute money in the future, thus it is considered more than a simple asset. In line with Nakamoto (2008), subjects who do not trust the banking system consider Bitcoin as an alternative monetary system. By taking as reference Italy, which is characterized by a low level of financial literacy, results remark the importance of increasing the level of financial literacy also among young investors.

CRediT authorship contribution statement

Alessandro Cascavilla: Conceptualization, Data curation, Writing – original draft, Writing – review & editing, Visualization, Investigation, Validation, Formal analysis, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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